Project Report

Course: DLBAIPNLP01 – Project: NLP

Task 1: Sentiment Analysis on Movie Reviews

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Abstract

This project explores the possibility of automatically determining the sentimental component of movie reviews using machine learning methods. The task involves classifying text-based reviews into one of two categories: positive or negative. To achieve this, we use the publicly available IMDb dataset which contains reviews with corresponding sentiment labels.

The study consists of analysis of the data structure, text preprocessing, transformation of text into numerical features, model training, and evaluation of performance on a separate test set. The project aims to compare a Naive Bayes classifier with logistic regression, as well as to research the impact of training set size on classification results.

The final version of the abstract will present the actual experimental results, key conclusions, and an assessment of the limitations of the developed system.

1. Introduction

1.1. Relevance of the research

As the volume of text-based reviews on the Internet increases, the automated analysis of user opinions has become a significant task in the field of natural language processing. Movie reviews contain subjective evaluation, emotional expressions, and various ways of expressing positive or negative opinions. The automated classification of such texts provides an opportunity to explore the application of machine learning to the analysis of human language.

At the same time, sentiment cannot always be determined only by individual words. Contradictions, mixed evaluations, and context can complicate the interpretation of a

text. Therefore, it is important not only to build a classifier but also to test it on new data for the model that was not used during the training process.

1.2. Objective of the work

The objective of this work is to develop and evaluate a system for the automated sentiment classification of movie reviews using machine learning methods.

1.3. Research tasks

To achieve this stated objective, the following tasks must be completed:

- Review the theoretical foundations of sentiment analysis in text.
- Load and explore the publicly available IMDb dataset.
- Preprocess the text reviews.
- Convert the texts into numerical features using the TF-IDF method.
- Train and compare two machine learning algorithms.
- Investigate the results using training sets of various sizes.
- Evaluate classification performance using appropriate metrics.
- Analyze incorrect predictions and model limitations.
- Formulate conclusions and prepare a reproducible software project.

1.4. Research questions

This work addresses three research questions:

- How does classification quality change as the volume of training data increases?
- Which of the selected models performs better under identical evaluation conditions?
- What characteristics of the texts might lead to misclassification?

Answers to these questions will be formulated after the experiments are done.

2. Materials and methods

2.1 The project is done using the Google Colab environment, which allows the execution of Python code in the cloud. This allows us to sequentially perform data preparation, model training, and result evaluation within a single document.

The code, explanations, and experimental results are stored in a notebook. A GitHub repository will be used to organize the software materials and provide access to the results.

2.2. Software tools

The pandas library is used for working with data in tables, while NumPy is for numerical operations. Tools from the scikit-learn library are used to build and evaluate machine learning models. The dataset is loaded using the datasets library.

Matplotlib is used to visualize the results. Text preprocessing involves a stemming algorithm.

3. Dataset

3.1. Data source and purpose

This study uses the IMDb Large Movie Review Dataset, introduced by Maas et al. (2011). The dataset was created for research on automated sentiment analysis. Inside of it, movie reviews are labeled as either positive or negative.

The use of pre-labeled data allows us to apply supervised learning. The model is provided with text samples along with their corresponding correct classes, based on which it derives classification rules.

3.2. Data structure and volume

The dataset consists of review text and an associated class label. A label value of 0 indicates a negative review, while a value of 1 indicates a positive one. The original dataset is divided into training and test sets.

The actual number of records and the class distribution were verified programmatically with loading the data.

3.3. Initial data quality check

Prior to training, the sample sizes, the presence of both classes, and the number of missing values in the texts and labels were checked. This step is essential for identifying potential issues that could affect processing and model evaluation.

The results of this check confirm that the data is ready for the following stages of the research.

```
Class distribution in training data:
label
Negative    12500
Positive    12500
Name: count, dtype: int64

Class distribution in testing data:
label
Negative    12500
Positive    12500
Name: count, dtype: int64

Missing values in training data:
text     0
label    0
dtype: int64

Missing values in testing data:
text     0
label    0
dtype: int64
```

Image 1. Initial data quality check

4. Text preprocessing

4.1. Cleaning data

Preprocessing is done to bring texts into a consistent format suitable for further analysis. At first, HTML entities are decoded, HTML tags are removed, and all characters are converted to lowercase.

These operations reduce the impact of technical differences between texts and help focus on their content. The cleaning process is performed using the same function for both the training and test datasets.

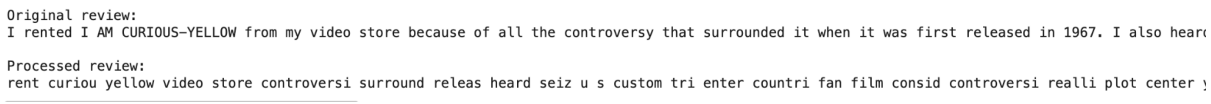


Image 2. Example of original review and its processed version

4.2. Tokenization and stemming

After cleaning, the texts are broken down into sequences of words. Then, stemming is applied to the words using the Porter algorithm, which reduces various word forms to a common root, which is the stem.

Although stemming can reduce the size of the feature vocabulary, it sometimes produces stems that are not valid words on their own. Therefore, its use has both advantages and limitations.

4.3. Handling stop words and negations

During text preparation, common English words are removed because they have limited discriminative value. Hovewer, negation words are retained, as they can significantly alter the sentiment of a statement.

This approach is designed to minimize the loss of important information during text cleaning. The processed texts are then used in the next stage to construct a numerical representation of the documents.

Processed review examples:

	text	processed_text	label
0	I rented I AM CURIOUS-YELLOW from my video sto...	rent curiou yellow video store controversi sur...	0
1	"I Am Curious: Yellow" is a risible and preten...	curiou yellow risibl pretenti steam pile doesn...	0
2	If only to avoid making this type of film in t...	avoid make type film futur film interest exper...	0

Number of empty processed reviews: 0

Image 3. Processed review examples with label 0 for negative and 1 for positive reviews

5. Converting text into numerical features

5.1. The TF-IDF method

The TF-IDF method is used to convert text documents into numerical features. It takes into account both how frequently a term is seen within an individual document and across the entire collection. Thus, words that help distinguish between documents can be assigned a higher weight than terms that appear too frequently.

The implementation uses both individual words and two-word sequences called bigrams. This allows the model to utilize not only information about single words but also specific short phrases that are significant for determining sentiment.

5.2. Constructing the feature matrix

Each review is represented as a numerical vector, and the collection of these vectors forms a feature matrix. The number of rows corresponds to the number of documents, while the number of columns corresponds to the number of generated text features.

The transformation parameters are fixed within the code to ensure the consistency of experiments. The transformer is fitted exclusively on the training texts and is later applied to the corresponding test set.

```
Number of documents: 1000
Number of features: 10000
Matrix shape: (1000, 10000)
```

Image 4. Outcome of the matrix construction

6. Model Construction and Training

6.1. Naive Bayes Classifier

A Multinomial Naive Bayes classifier is used for the classification of reviews. The algorithm estimates the probability that a document belongs to a specific class based on textual features. It is a common baseline method for text classification tasks.

6.2. Logistic Regression

The second algorithm employed is logistic regression. In this task, it serves as a binary classification model trained on a numerical representation of the texts. Its performance is compared against that of the Naive Bayes classifier under identical evaluation conditions.

6.3. Experiments with Varying Training Data Volumes

To investigate the impact of the number of training examples, four different training set sizes are used: 1,000, 5,000, 10,000, and 25,000 reviews. For each size, balanced subsets are created, containing an equal number of positive and negative examples.

Both models are evaluated on the same test set. This approach allows for the comparison of results across different models and varying volumes of training data. The resulting metrics are used for further analysis of classification performance.

7. Model training and experimental design

The task of the experimental phase of this study was to develop and compare two machine learning algorithms for the automated sentiment classification of movie

reviews. They are Multinomial Naive Bayes and Logistic Regression. The task was designed as a binary classification problem, where each review had to be assigned to one of two classes: positive or negative. The IMDb dataset, commonly used in research on automated sentiment analysis (Maas et al., 2011), was utilized for the experiment.

Multinomial Naive Bayes is based on a probabilistic approach to classification, and Logistic Regression estimates the probability that a given instance belongs to a specific class. Developing both algorithms allowed us to compare the results by two common text classification approaches, while having a consistent method of data representation (Lane et al., 2019). To investigate the impact of the number of training examples, four training set sizes were used: 1,000, 5,000, 10,000, and 25,000 reviews. For each size, we created subsamples containing an equal number of positive and negative examples. Both models were evaluated on the same test set. This approach allows for the comparison of results across models and across different volumes of training data.

Classification quality was measured using accuracy, precision, recall, and F1 score. Additionally, the execution time for each experiment was recorded. The use of multiple metrics provides a better view of classification quality, as each metric characterizes a specific aspect of the prediction results. This experimental design made it possible to examine the influence of training set size and to compare the performance of the two algorithms under identical testing conditions.

Model comparison:

	Training Set Size	Model	Accuracy	Precision	Recall	F1-score	Time (seconds)
0	1000	Logistic Regression	0.81552	0.807261	0.82896	0.817967	0.73
1	5000	Logistic Regression	0.85076	0.845318	0.85864	0.851927	0.56
2	10000	Logistic Regression	0.86628	0.861166	0.87336	0.867220	0.72
3	25000	Logistic Regression	0.88064	0.879124	0.88264	0.880878	0.55
4	1000	Naive Bayes	0.81068	0.839615	0.76808	0.802256	0.02
5	5000	Naive Bayes	0.83096	0.868257	0.78032	0.821943	0.03
6	10000	Naive Bayes	0.84184	0.862364	0.81352	0.837230	0.03
7	25000	Naive Bayes	0.85008	0.866868	0.82720	0.846570	0.04

Image 5. Model comparison

8. Visualization and analysis of experimental results

To research how classification quality depends on the size of the training data, we created graphs illustrating changes in Accuracy and F1 Score for the Multinomial Naive Bayes and Logistic Regression algorithms. The horizontal axis shows the number of reviews in the training set, and the vertical axis shows the corresponding value of the evaluated metric. Separate lines represent the results for each algorithm, allowing for a comparison of their performance across different training set sizes.

Accuracy represents the overall proportion of correctly classified instances. And the F1 Score combines Precision and Recall using their harmonic mean. Consideration of both metrics makes it possible to assess the overall correctness of predictions and estimate the balance between the precision of positive predictions and the model's ability to detect positive instances.

An increase in metric values may indicate that the model would benefit from additional training examples. At the same time, if the metrics don’t increase consistently, it does not necessarily mean an error. This is because the nature of the relationship must be determined based on the actual experimental results.

According to the findings, the highest Accuracy value was 0.88, achieved by the Logistic Regression model with a training set size of 25000. The highest F1 Score was 0.88, corresponding to the configuration of the Logistic Regression model with the set size of 25000. A comparison of the graphs showed that as the training set size increased, the metrics also increased for both models and metrics.

Thus, the visualization made it possible to clearly compare the model results and investigate the impact of the training set size on classification quality. These observations set the basis for a more detailed comparison of the algorithms.

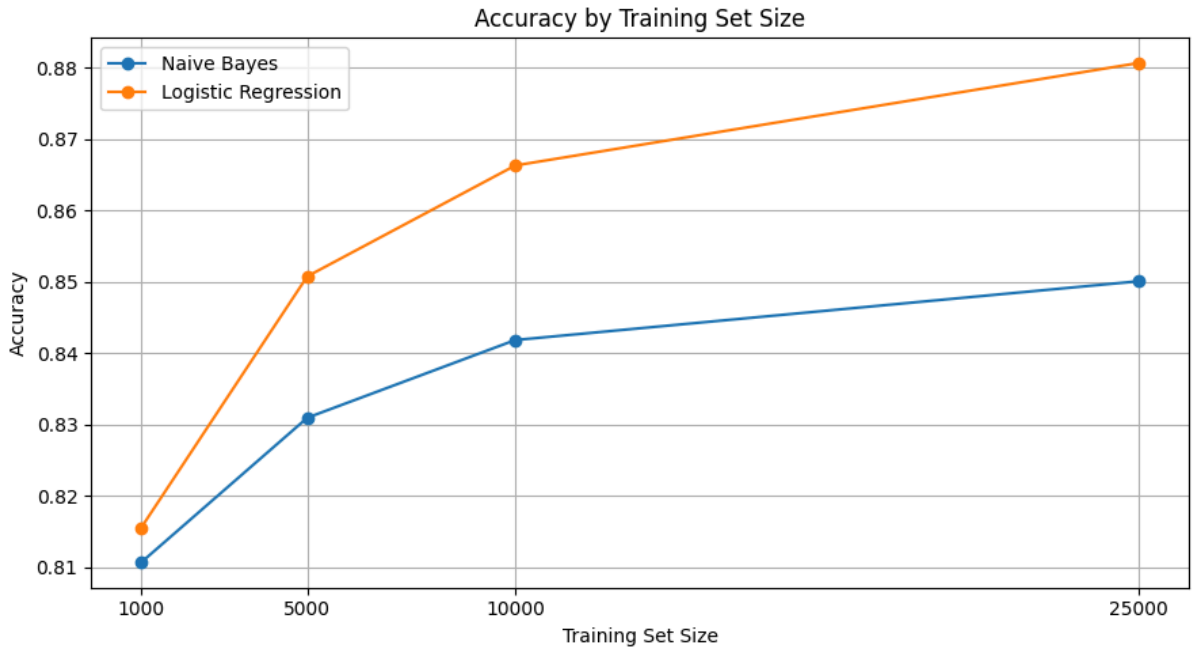


Image 6. Comparison of accuracy of two models

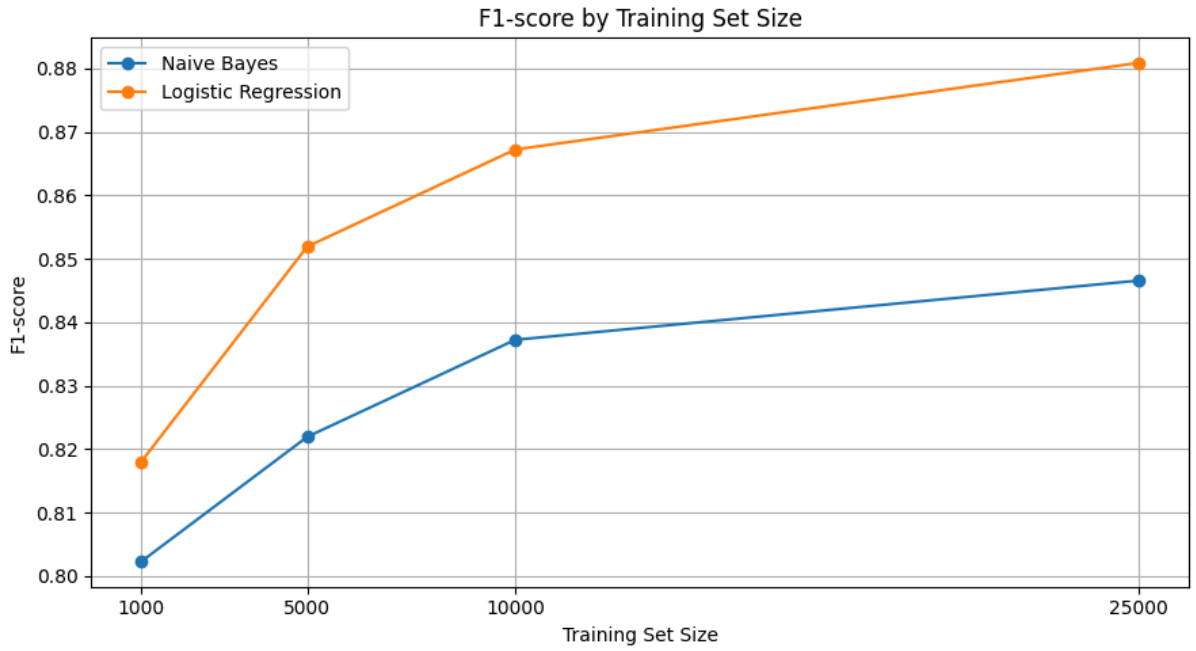


Image 7. Comparison of F1-score of two models

9. Comparative evaluation of machine learning models

To determine the relative effectiveness of Multinomial Naive Bayes and Logistic Regression, we conducted a comparison based on Accuracy, Precision, Recall, F1 Score, and the experiment execution time.

Execution time was included in the comparison as an additional indicator of computational efficiency. When selecting a model, it is important to consider not only prediction quality but also the resources required for training. However, since the recorded time depends on the execution environment, it should be viewed as the outcome of a specific experiment rather than a universal characteristic of the algorithm's speed.

Based on the comparison, the highest F1 Score for Multinomial Naive Bayes was 0.846570, achieved with a training set size of 25000 reviews. For Logistic Regression, the corresponding value was 0.880878, with a training set size of 25000 reviews. The best Accuracy, Precision, and Recall values for each configuration are presented in the table of experimental results. Based on the results obtained, the

Logistic Regression model was selected for further analysis, as it demonstrated the highest F1 score, higher accuracy and precision, which are balanced with the slightly higher execution time. This selection applies specifically to the conditions of the conducted study. It does not imply that this model would be the best choice for any text dataset, as classification results depend on corpus characteristics, text representation, and algorithm parameters (Lane et al., 2019).

	Training Set Size	Model	Accuracy	Precision	Recall	F1-score	Time (seconds)
0	1000	Logistic Regression	0.81552	0.807261	0.82896	0.817967	0.73
1	1000	Naive Bayes	0.81068	0.839615	0.76808	0.802256	0.02
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7	25000	Naive Bayes	0.85008	0.866868	0.82720	0.846570	0.04

Best F1-score for each model:

	Training Set Size	Model	Accuracy	Precision	Recall	F1-score	Time (seconds)
0	25000	Logistic Regression	0.88064	0.879124	0.88264	0.880878	0.55
1	25000	Naive Bayes	0.85008	0.866868	0.82720	0.846570	0.04

Image 8. Comparative evaluation of machine learning models

10. Evaluating the selected model using a confusion matrix

After comparing the results of two classifiers, Logistic Regression was selected for further analysis. The confusion matrix displays the number of correctly and incorrectly classified positive and negative reviews. The obtained results are used for a more detailed analysis of the Logistic Regression model's performance on the test dataset.

```
Selected model: Logistic Regression
Confusion matrix:
[[10983  1517]
 [ 1467 11033]]
```

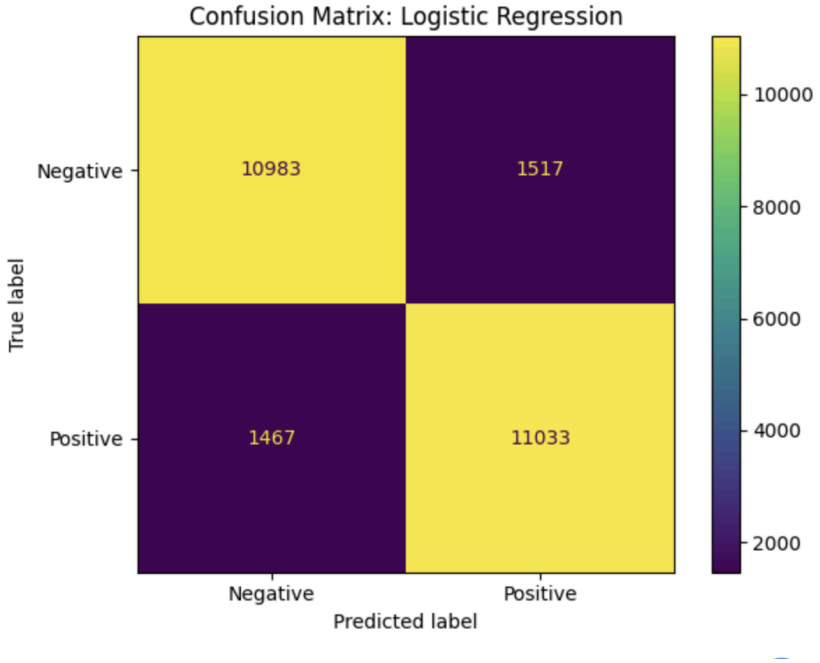


Image 9. Confusion matrix: logistic regression

11. Analysis of misclassifications

While the confusion matrix provides the total number of errors, we must also examine the specific reviews where the model's prediction did not match the actual label. For instance, a model might misclassify a review in which the author praises the acting but criticizes the plot. Other challenging cases can be sarcasm, irony, negation, and statements whose meaning depends on the context. These are typical difficulties in sentiment analysis (Lane et al., 2019; Paramesha et al., 2023).

To gain a deeper understanding of the Logistic Regression model's performance, we analyzed reviews where the predicted label did not match the actual label from the test set. For each such instance, the original text, the correct class, and the model's prediction were recorded.

The number of misclassified instances and their content are used to interpret the model's results. However, since individual examples cannot fully explain the reasons behind every error, these findings are presented as illustrative cases.

```
Selected model: Logistic Regression
Total test reviews: 25000
Misclassified reviews: 2984
Misclassification rate: 0.1194
```

	text	processed_text	Correct Label	Predicted Label
4	First off let me say, If you haven't enjoyed a...	let say haven t enjoy van damm movi bloodsport...	0	1
6	Isaac Florentine has made some of the best wes...	isaac florentin best western martial art actio...	0	1
11	Blind Date (Columbia Pictures, 1934), was a de...	blind date columbia pictur decent film issu fi...	0	1
20	Low budget horror movie. If you don't raise yo...	low budget horror movi don t rais expect high ...	0	1
32	I'm the type of guy who loves hood movies from...	m type guy love hood movi new jack citi babi b...	0	1
36	Beware, My Lovely (1952) Dir: Harry Horner <br...	bewar love dir harri horner product filmmak rk...	0	1
41	Widow hires a psychopath as a handyman. Sloppy...	widow hire psychopath handyman sloppi film noi...	0	1
46	Okay, so it was never going to change the worl...	okay never go chang world bomb box offic honki...	0	1
47	A sprawling, overambitious, plotless comedy th...	sprawl overambiti plotless comedi no dramat ce...	0	1
61	This film features two of my favorite guilty p...	film featur favorit guilti pleasur sure effect...	0	1

Image 10. Analysis of misclassifications

12. Model testing on new text examples

Following the evaluation of the model on the test set, we performed an additional check using four new text examples. The objective of this stage was to demonstrate the application of the trained classifier in determining the positive or negative sentiment of reviews.

Predictions were generated using Logistic Regression and a TF-IDF vectorizer, both trained on a dataset of 25,000 reviews. The new texts went through the same preprocessing procedure as the training data. So, they were then converted into numerical features and fed into the classifier.

The results of this experiment illustrate how the system can classify individual new reviews. Given the small number of examples, this test serves a demonstrative purpose and does not replace quantitative evaluation on a dedicated test set.

```
Review: This movie was wonderful and deeply moving.
Predicted sentiment: Positive

Review: The film was boring and a complete disappointment.
Predicted sentiment: Negative

Review: The acting was excellent but the story was weak.
Predicted sentiment: Positive

Review: I would definitely recommend this movie to everyone.
Predicted sentiment: Positive
```

Image 11. Model performance on new individual text examples

13. Study Limitations

The study has several limitations that should be considered when interpreting the results. First, TF-IDF was used for text representation. While this method converts words and phrases into numerical features, it does not capture the full context or the semantic relationships between words. Thus, the classifier may struggle to detect sarcasm, irony, negation, and mixed sentiment.

Second, the results are dependent on the characteristics of the dataset, the preprocessing procedures, and the model parameters. Performance metrics derived from movie reviews cannot be automatically generalized to other text types or domains. Research on sentiment analysis across various domains further shows the importance of accounting for data-specific characteristics and the chosen classification approach (Maas et al., 2011; Paramesha et al., 2023).

Third, this project compared Multinomial Naïve Bayes and Logistic Regression. The selection of Logistic Regression is based specifically on the results of these

experiments and does not imply that it will necessarily outperform other algorithms under all conditions.

In summary, while the study's findings provide an assessment of the effectiveness of the selected methods for the binary classification of movie reviews within the context of the dataset used, they should be interpreted with these limitations in mind.

14. Conclusion

Within the scope of this project, a binary sentiment classification system for movie reviews was developed. The TF-IDF method was used to transform text data into numerical features, and two machine learning algorithms were implemented for classification. They were Multinomial Naive Bayes and Logistic Regression.

Experiments were conducted using four different training set sizes: 1,000, 5,000, 10,000, and 25,000 reviews. Model performance was compared using Accuracy, Precision, Recall, and F1-score. Both models were evaluated on the same test set, ensuring that their results could be compared under identical conditions.

Based on the experimental results, Logistic Regression was selected for further analysis, as it consistently outperformed Multinomial Naive Bayes across the tested configurations. The final numerical results are presented in the model comparison table.

Further evaluation of the Logistic Regression model included constructing a confusion matrix, analyzing misclassified reviews, and testing the model on new text samples. These steps made it possible not only to assess classification quality but also to examine instances where the model failed to correctly identify the sentiment.

Overall, the results demonstrate the effectiveness of using TF-IDF in combination with classical machine learning algorithms for the binary sentiment classification of

movie reviews. At the same time, the system's performance depends on data characteristics, selected parameters, and the complexity of linguistic structures. Therefore, the results obtained should be interpreted within the context of the experiment conducted (Jurafsky & Martin, 2013; Lane et al., 2019; Maas et al., 2011).

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16. Reference list

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17. Dataset reference

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